

- **University Campus**



T E A M

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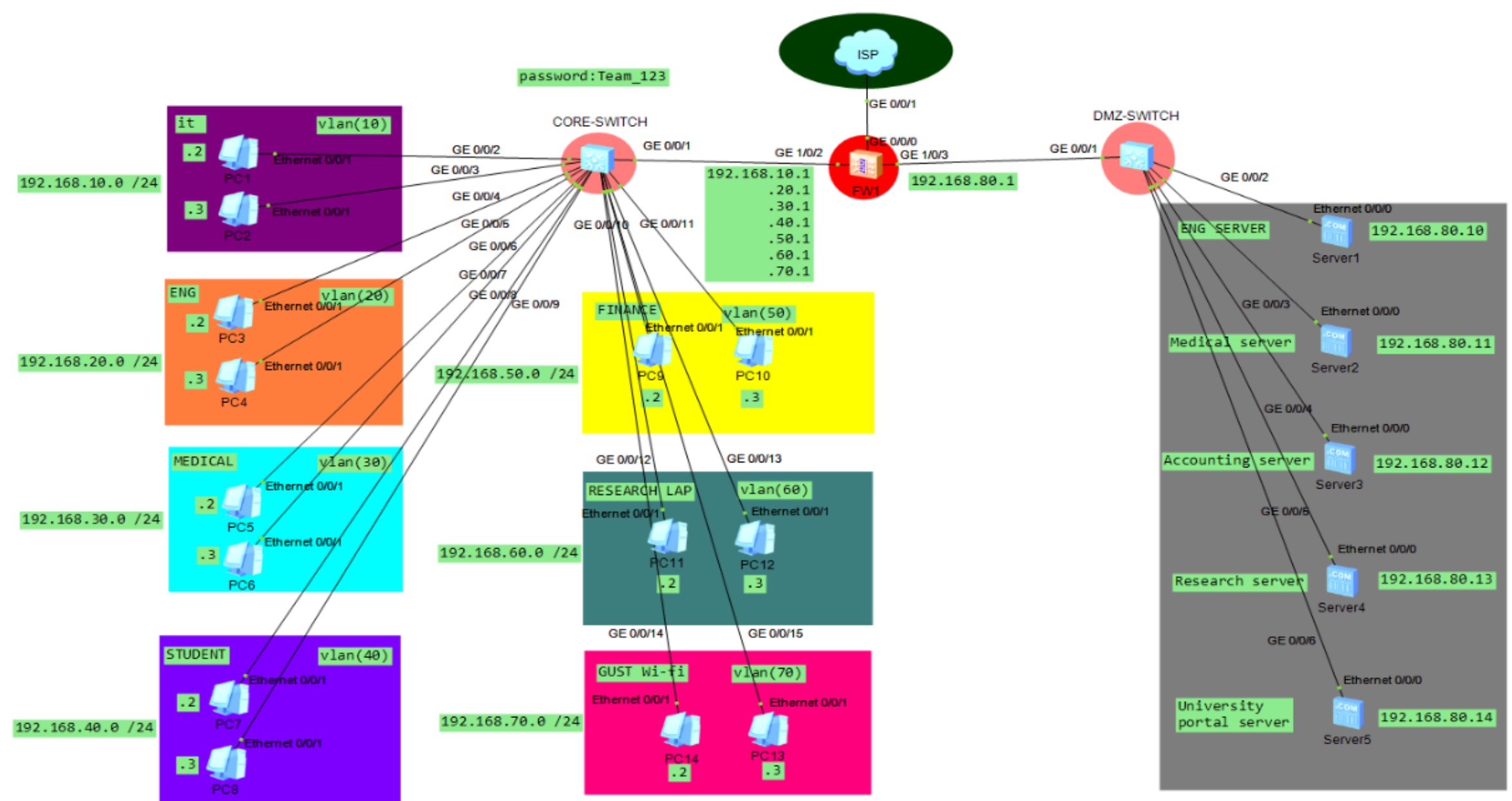
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● Project Scenario

- **We are building a secure network for a university campus. The goal is to separate the different departments and control their access.**

- 1. IT Full Control: The IT department can access all servers to manage the network.**
- 2. Department Isolation: Every faculty (like Medical or Engineering) has its own safe network. They can only open their own specific server.**
- 3. Time Rules: Engineering and Research students can only use their servers during lab time (from 08:00 AM to 06:00 PM).**
- 4. Internet Filter: Students cannot open social media or games. Medical staff can only open medical websites.**
- 5. Guest Internet: Guest Wi-Fi has a speed limit to keep the network fast for others.**



1. Firewall Network Interfaces (GUI)

 **Huawei**
USG6000V1-ENSP

 Dashboard  Monitor  Policy  Object  Network  System

Current User: **admin** [Commit](#) [Save](#) [Help](#) [About](#) [Change Password](#) [Logout](#)

Interface

Interface

Link Interface

Link Interface Group

Interface Pair

Zone

VXLAN

DNS

DHCP Server

Route

IPSec

L2TP

L2TP over IPSec

GRE

SSL VPN

SACG

Interface List

[+ Add](#) [X Delete](#) [Refresh](#) [Search](#) [Clear Search Condition](#)

Interface Name	Zone	IP Address	Connectio...	VLAN/VX...	Mode	State			Ena...	Edit
						Physical	IPv4	IPv6		
GE1/0/1	untrust(public)	---	DHCP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	☒	
GE1/0/2	-- NONE --(public)	---	Static IP (IPv4) Static IP (IPv6)		Routing	↑	↓	↓	☒	
☐ GE1/0/2.10	z.it(public)	192.168.10.1	Static IP (IPv4) Static IP (IPv6)	10	Routing	↑	↑	↓	☒	
☐ GE1/0/2.20	z.eng(public)	192.168.20.1	Static IP (IPv4) Static IP (IPv6)	20	Routing	↑	↑	↓	☒	
☐ GE1/0/2.30	z.medical(public)	192.168.30.1	Static IP (IPv4) Static IP (IPv6)	30	Routing	↑	↑	↓	☒	
☐ GE1/0/2.40	z.student(public)	192.168.40.1	Static IP (IPv4) Static IP (IPv6)	40	Routing	↑	↑	↓	☒	
☐ GE1/0/2.50	z.finance(public)	192.168.50.1	Static IP (IPv4) Static IP (IPv6)	50	Routing	↑	↑	↓	☒	
☐ GE1/0/2.60	z.research.lap(public)	192.168.60.1	Static IP (IPv4) Static IP (IPv6)	60	Routing	↑	↑	↓	☒	
☐ GE1/0/2.70	trust(public)	192.168.70.1	Static IP (IPv4) Static IP (IPv6)	70	Routing	↑	↑	↓	☒	
GE1/0/3	dmz(public)	192.168.80.1	Static IP (IPv4) Static IP (IPv6)		Routing	↑	↑	↓	☒	
GE1/0/4	---	---	Static IP (IPv4)		Routing	↓	↓	↓	☒	

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[CLI Console](#)

2. Firewall Security Policies (GUI)

Huawei FW1

Not secure | https://192.168.0.1:8443/default.html

To get future Google Chrome updates, you'll need Windows 10 or later. This computer is using Windows 7.

HUAWEI

Huawei

USG6000V1-ENSP

Dashboard

Monitor

Policy

Object

Network

System

Current User: admin

Commit

Save

Help

About

Change Password

Logout

Security Policy

Security Policy

NAT Policy

Server Load Balancing

Bandwidth Management

Quota Control Policy

Proxy Policy

Security Protection

ASPF Configuration

Security Policy List

+ Add Security Policy

+ Add Security Policy Group

✖ Delete

📄 Copy

↔ Move

📄 Insert

📄 Export

📄 Reset All Statistics

📄 Enable

📄 Disable

📄 Customize

📄 Expand

📄 Collapse

🔄 Refresh

🔍 Match Query

🗑 Clear Match Query

🔍 Enter a keyword.

+ Add Filter

Index	Name	Descri...	Tag	VLAN ID	Sourc...	Destin...	Sourc...	Destin...	User	Service	Applic...	Sched...	Action	Conte...	Hits	Enable	Edit
☐ 1	it-to-dmz			any	🌐 z.it	🌐 dmz	any	any	any	any	any	any	Permit		0	Reset	👉
☐ 2	eng-to-server			any	🌐 z.eng	🌐 dmz	any	192.16...	any	any	any	lap-time	Permit		0	Reset	👉
☐ 3	med-to-server			any	🌐 z.medic...	🌐 dmz	any	192.16...	any	any	any	any	Permit		0	Reset	👉
☐ 4	fin-to-server			any	🌐 z.financi...	🌐 dmz	any	192.16...	any	any	any	any	Permit		0	Reset	👉
☐ 5	res-to-server			any	🌐 z.resear...	🌐 dmz	any	192.16...	any	any	any	any	Permit		0	Reset	👉
☐ 6	stu-to-server			any	🌐 z.studer...	🌐 dmz	any	192.16...	any	any	any	any	Permit		0	Reset	👉
☐ 7	block-dtu-soc...			any	🌐 z.studer...	🌐 untrust	any	8.8.8.8/...	any	any	any	any	Deny		0	Reset	👉
☐ 8	stu-to-interne...			any	🌐 z.studer...	🌐 untrust	any	any	any	any	any	any	Permit		0	Reset	👉
☐ 9	block-med-int...			any	🌐 z.medic...	🌐 untrust	any	any	any	any	any	any	Deny		0	Reset	👉
☐ 10	all-to-internet			any	🌐 z.it 🌐 z.eng 🌐 z.financi...	🌐 untrust	any	any	any	any	any	any	Permit		0	Reset	👉
11	default	This is ...		any	any	any	any	any	any	any	any	any	Deny		7	Reset	👉

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CLI Console

• NAT Policy

 **Huawei**
USG6000V1-ENSP

 Dashboard  Monitor  Policy  Object  Network  System

Current User: **admin** [Commit](#) [Save](#) [Help](#) [About](#) [Change Password](#) [Logout](#)

[-]  Security Policy

[-]  NAT Policy

[-]  Server Mapping

[-]  Server Load Balancing

[-]  Bandwidth Management

[-]  Quota Control Policy

[-]  Proxy Policy

[-]  Security Protection

[-]  ASPF Configuration

NAT Policy

Source Translation Address Pool

Destination Translation Address Pool

NAT Policy List

 Add  Delete  Copy  Move  Insert  Reset All Statistics  Enable  Disable  Refresh

 Add Filter 

☐ Name	Description	Tag	NAT Type	NAT Mode	Schedule
☐ internet-access			NAT	Source address translation	any
default	This is the default rule			No translation	any

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[CLI Console](#)

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CLI Configuration — Core Switch

VERIFICATION: DISPLAY CURRENT-CONFIGURATION

```
system-view
sysname CORE-SWITCH
vlan batch 10 20 30 40 50 60 70
interface GigabitEthernet 0/0/1
port link-type trunk
port trunk allow-pass vlan 10 20 30 40 50 60 70
quit
interface GigabitEthernet 0/0/2
port link-type access
port default vlan 10
quit
interface GigabitEthernet 0/0/3
port link-type access
port default vlan 10
quit
interface GigabitEthernet 0/0/4
port link-type access
port default vlan 20
quit
interface GigabitEthernet 0/0/5
port link-type access
port default vlan 20
quit
interface GigabitEthernet 0/0/6
port link-type access
port default vlan 30
quit
interface GigabitEthernet 0/0/7
port link-type access
port default vlan 30
quit
interface GigabitEthernet 0/0/8
port link-type access
port default vlan 40
quit
interface GigabitEthernet 0/0/9
port link-type access
port default vlan 40
quit
interface GigabitEthernet 0/0/10
port link-type access
port default vlan 50
quit
interface GigabitEthernet 0/0/11
port link-type access
port default vlan 50
quit
interface GigabitEthernet 0/0/12
port link-type access
port default vlan 60
quit
interface GigabitEthernet 0/0/13
port link-type access
port default vlan 60
quit
interface GigabitEthernet 0/0/14
port link-type access
port default vlan 70
quit
interface GigabitEthernet 0/0/15
port link-type access
port default vlan 70
quit
```

Firewall CLI — Interfaces & Security Zones

Interface Setup

```
system-view
sysname FW1

# Interfaces Setup
interface GigabitEthernet 1/0/1
ip address dhcp-alloc
quit
interface GigabitEthernet 1/0/3
ip address 192.168.80.1 255.255.255.0
quit
interface GigabitEthernet 1/0/2.10
vlan-type dot1q 10
ip address 192.168.10.1 255.255.255.0
quit
interface GigabitEthernet 1/0/2.20
vlan-type dot1q 20
ip address 192.168.20.1 255.255.255.0
quit
interface GigabitEthernet 1/0/2.30
vlan-type dot1q 30
ip address 192.168.30.1 255.255.255.0
quit
interface GigabitEthernet 1/0/2.40
vlan-type dot1q 40
ip address 192.168.40.1 255.255.255.0
quit
interface GigabitEthernet 1/0/2.50
vlan-type dot1q 50
ip address 192.168.50.1 255.255.255.0
quit
interface GigabitEthernet 1/0/2.60
vlan-type dot1q 60
ip address 192.168.60.1 255.255.255.0
quit
interface GigabitEthernet 1/0/2.70
vlan-type dot1q 70
ip address 192.168.70.1 255.255.255.0
quit
```

Security Zones Setup

```
firewall zone untrust
add interface GigabitEthernet 1/0/1
quit
firewall zone dmz
add interface GigabitEthernet 1/0/3
quit
firewall zone trust
add interface GigabitEthernet 1/0/2.70
quit
firewall zone name z.it
set priority 75
add interface GigabitEthernet 1/0/2.10
quit
firewall zone name z.eng
set priority 70
add interface GigabitEthernet 1/0/2.20
quit
firewall zone name z.medical
set priority 65
add interface GigabitEthernet 1/0/2.30
quit
firewall zone name z.student
set priority 60
add interface GigabitEthernet 1/0/2.40
quit
firewall zone name z.finance
set priority 55
add interface GigabitEthernet 1/0/2.50
quit
firewall zone name z.research.lap
set priority 45
add interface GigabitEthernet 1/0/2.60
quit
```


UTM Profiles & Time-Range

Time-Range & URL Filtering

```
time-range LAB-TIME 08:00 to 18:00 daily
```

```
profile type url-filter name MEDICAL_ONLY  
default action block  
quit
```

```
profile type url-filter name STUDENTS_RESTRICTED  
category add Pre-defined/Social_Networking action  
block  
category add Pre-defined/Games action block  
default action allow  
quit
```

QoS Profile & Traffic Policy

```
profile type qos name GUEST_LIMIT  
bandwidth maximum-bandwidth 5000  
quit
```

```
traffic-policy  
rule name LIMIT_GUEST_TRAFFIC  
source-zone trust  
destination-zone untrust  
action qos  
profile GUEST_LIMIT  
quit  
quit
```

Security Policies — DMZ Access Rules

Each department has a dedicated rule permitting access to its assigned DMZ server with session logging enabled.

```
security-policy
rule name IT_TO_DMZ
source-zone z.it
destination-zone dmz
action permit
session logging enable
quit
rule name ENG_TO_SERVER
source-zone z.eng
destination-zone dmz
destination-address 192.168.80.10 mask 255.255.255.255
time-range LAB-TIME
action permit
session logging enable
quit
rule name MED_TO_SERVER
source-zone z.medical
destination-zone dmz
destination-address 192.168.80.11 mask 255.255.255.255
action permit
session logging enable
quit
rule name FIN_TO_SERVER
source-zone z.finance
destination-zone dmz
destination-address 192.168.80.12 mask 255.255.255.255
action permit
session logging enable
quit
rule name RES_TO_SERVER
source-zone z.research.lap
destination-zone dmz
destination-address 192.168.80.13 mask 255.255.255.255
time-range LAB-TIME
action permit
session logging enable
quit
rule name STU_TO_SERVER
source-zone z.student
destination-zone dmz
destination-address 192.168.80.14 mask 255.255.255.255
action permit
session logging enable
quit
```

Security Policies — Internet Access Rules

```
rule name STU_TO_INTERNET
source-zone z.student
destination-zone untrust
action permit
profile url-filter STUDENTS_RESTRICTED
session logging enable
quit
rule name MED_TO_INTERNET
source-zone z.medical
destination-zone untrust
action permit
profile url-filter MEDICAL_ONLY
session logging enable
quit
rule name ALL_TO_INTERNET
source-zone z.it
source-zone z.eng
source-zone z.finance
source-zone trust
destination-zone untrust
action permit
session logging enable
quit
quit
```

Student Internet

Social networking and gaming categories blocked; all other traffic permitted.

Medical Internet

Default block — only explicitly whitelisted URLs are accessible.

Full Internet Access

IT, Engineering, Finance, and Guest Wi-Fi zones with unrestricted outbound access.

NAT Policy

SourceNATwith easy-iptranslates all internal zone addresses to the firewall's external interface IP, enabling internet access while preserving the internal addressing scheme.

```
nat-policy
rule name INTERNET_ACCESS
source-zone z.it
source-zone z.eng
source-zone z.medical
source-zone z.finance
source-zone z.student
source-zone trust
destination-zone untrust
action source-nat
easy-ip
quit
quit
```

Source Zones

All six internal departments plus Guest Wi-Fi (trust zone) are covered by this single NAT rule.

easy-ip Mode

Dynamically uses the IP of GE 1/0/1 (DHCP-assigned) as the translation address — no static pool required.

Destination Zone

NAT is triggered only for traffic destined to untrust, keeping internal traffic untranslated.

• Network Interfaces & IP Addressing

Eight isolated zones are mapped to dedicated VLANs and firewall sub-interfaces, enforcing segmentation across the campus network.

Interface	Zone / Department	VLAN	IP Address	Subnet Mask
GE 1/0/1	untrust (Internet)	—	DHCP Allocated	—
GE 1/0/3	dmz (Servers)	—	192.168.80.1	255.255.255.0
GE 1/0/2.10	z.it (IT Dept)	10	192.168.10.1	255.255.255.0
GE 1/0/2.20	z.eng (Engineering)	20	192.168.20.1	255.255.255.0
GE 1/0/2.30	z.medical (Medical)	30	192.168.30.1	255.255.255.0
GE 1/0/2.40	z.student (Students)	40	192.168.40.1	255.255.255.0
GE 1/0/2.50	z.finance (Finance)	50	192.168.50.1	255.255.255.0
GE 1/0/2.60	z.research.lap (Research)	60	192.168.60.1	255.255.255.0
GE 1/0/2.70	trust (Guest Wi-Fi)	70	192.168.70.1	255.255.255.0

Thanks